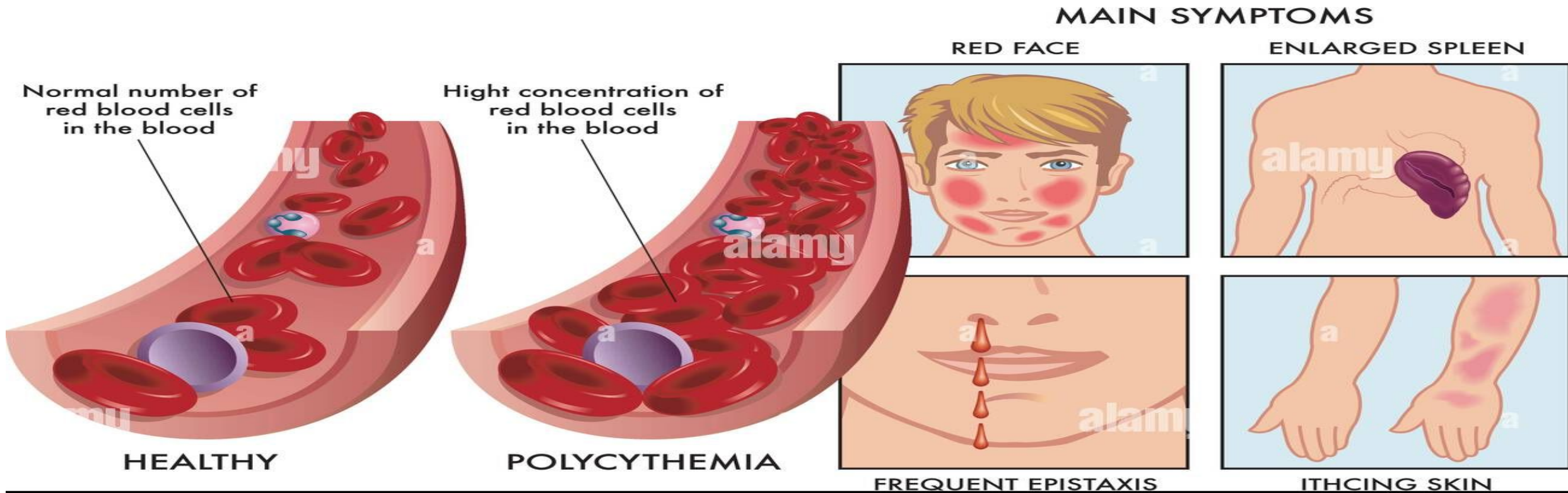


POLYCYTHEMIA



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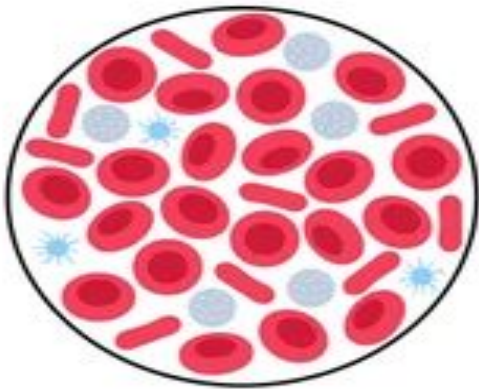
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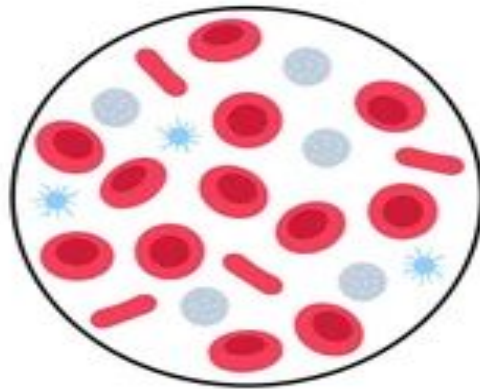
Polycythemia

Polycythemia is a blood disorder characterized by an abnormally high concentration of red blood cells (RBCs), hemoglobin, or hematocrit.

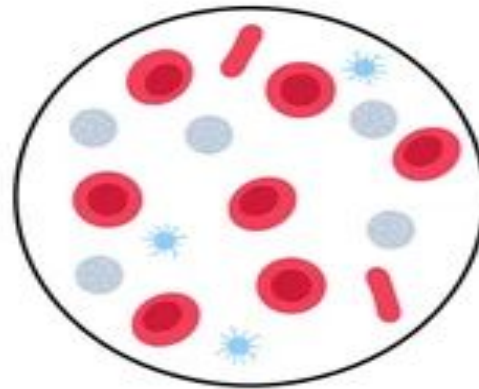
This increase thickens the blood, making it more viscous and slowing its flow, which can lead to serious complications like blood clots, heart attack, or stroke.



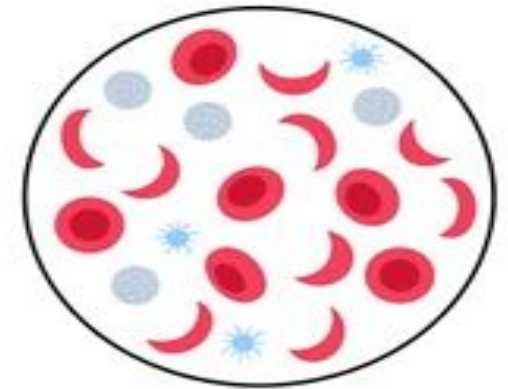
Polycythemia



Normal blood



Anemia



Sickle cell anemia

Plasma:

- Water, proteins, nutrients, hormones, etc.

Buffy coat:

- White blood cells, platelets

Hematocrit:

- Red blood cells

Normal Blood:

♀ 37%–47% hematocrit
♂ 42%–52% hematocrit

Anemia:

Depressed
hematocrit %

Polycythemia:

Elevated
hematocrit %

Classification

- **Relative Polycythemia:**

Apparent increase in RBCs due to decreased plasma volume (e.g., from dehydration).

- **Absolute Polycythemia:**

Actual increase in total RBC mass.

- **Primary (Polycythemia Vera):**

- A bone marrow disorder, often caused by a mutation in the JAK2 gene.

- **Secondary:**

- Caused by external factors like high altitude, smoking, or chronic lung/heart diseases that trigger excessive erythropoietin (EPO) production.

Types of Polycythemia

It occurs in both physiological and pathological conditions.

When it occurs in physiological conditions it is called physiological polycythemia.

A common type of secondary polycythemia, called physiological polycythemia, occurs in those who live at altitudes of 14,000 to 17,000 feet, where the atmospheric oxygen is very low. The blood count is generally 6 to 7 million/mm³, which allows these people to perform reasonably high

The increase in number during this condition is marginal and temporary. It occurs in the following conditions:

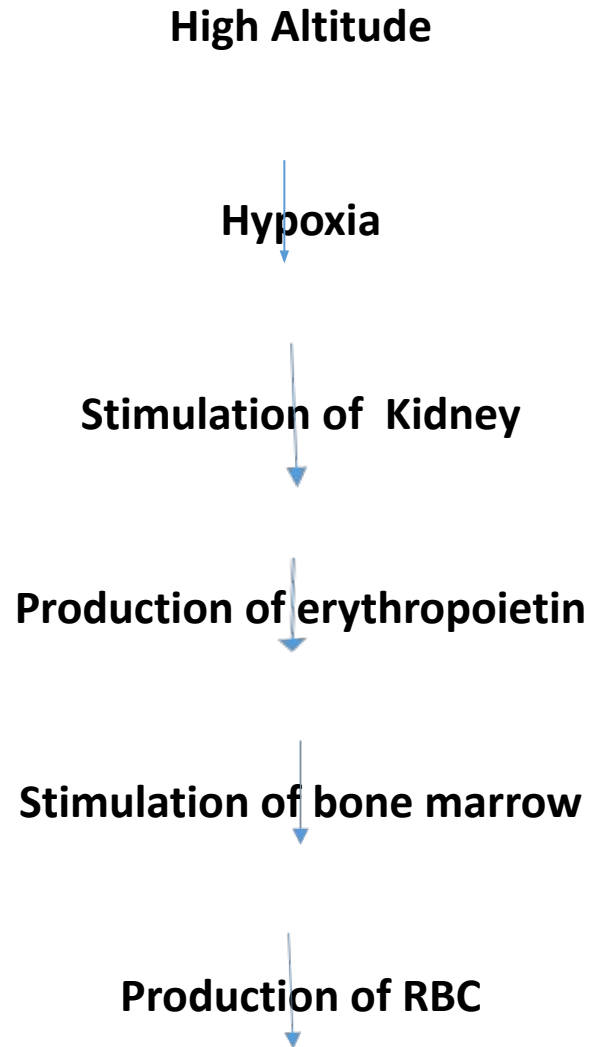
1- Age

- At birth, the RBC count is 8 to 10 million/cu mm of blood. The count decreases within 10 days after birth due to destruction of RBCs causing physiological jaundice in some newborn babies. However, in infants and growing children, the cell count is more than the value in adults.

2-Sex

- Before puberty and after menopause in females the RBC count is similar to that in males. During reproductive period of females, the count is less than that of males, (4.5 million/cu mm).

3-High altitude Inhabitants of mountains (above 10,000 feet from mean sea level) have an increased RBC count of more than 7 million/cu mm. It is due to hypoxia (decreased oxygen supply to tissues) in high altitude. Hypoxia stimulates kidney to secrete a hormone called erythropoietin. The erythropoietin in turn stimulates the bone marrow to produce more RBCs .



4-Muscular exercise

There is a temporary increase in RBC count after exercise. It is because of mild hypoxia and contraction of spleen. Spleen stores RBCs. Hypoxia increases the sympathetic activity resulting in secretion of adrenaline from adrenal medulla. Adrenaline contracts spleen and RBCs are released into blood .

Emotional condition and muscular exercise

Increase sympathetic activity



Increased secretion of adrenaline



Contraction of Spleen



Release of RBCs



5-Emotional conditions

RBC count increases during the emotional conditions such as anxiety. It is because of increase in the sympathetic activity as in the case of muscular exercise.

6-Increased environmental temperature

Increase in atmospheric temperature increases RBC count. Generally increased temperature increases all the activities in the body including production of RBCs.

7-After meals

There is a slight increase in the RBC count after taking meals. It is because of need for more oxygen for metabolic activities.

Pathological Polycythemia

Pathological polycythemia is the abnormal increase in the RBC count. Red cell count increases above 7 million/ cu mm of the blood. Polycythemia is of two types, the

1-Primary polycythemia and 2-secondary polycythemia.

•Primary Polycythemia – Polycythemia Vera

Primary polycythemia is also known as polycythemia vera.

It is a disease characterized by persistent increase in RBC count above 14 million/cu mm of blood. This is always associated with increased white blood cell count above 24,000/cu mm of blood. Polycythemia vera occurs in **myeloproliferative disorders like malignancy of red bone marrow.**

Polycythemia Vera is caused by a genetic aberration in the hemocytoblastic cells that produce the blood cells. The blast cells no longer stop producing RBCs when too many cells are already present. This causes excess production of RBCs. It usually causes excess production of white blood cells and platelets as well.

In polycythemia vera, not only does the hematocrit increase, but the total blood volume also increases, sometimes to almost twice normal. As a result, the entire vascular system becomes intensely engorged. Also, many blood capillaries become plugged by the viscous blood;

The viscosity of the blood in polycythemia vera sometimes increases from the normal of 3 times the viscosity of water to 10 times that of water

Secondary Polycythemia

- This is secondary to some of the pathological conditions (diseases) such as:
 - 1. Respiratory disorders like emphysema.
 - 2. Congenital heart disease.
 - 3. Ayerza's disease (condition associated with hypertrophy of right ventricle and obstruction of blood flow to lungs).
 - 4. Chronic carbon monoxide poisoning.
 - 5. Poisoning by chemicals like phosphorus and arsenic.
 - 6. Repeated mild hemorrhages.
- All these conditions lead to hypoxia which stimulates the release of erythropoietin. Erythropoietin stimulates the bone marrow resulting in increased RBC count

Symptoms

Headache, dizziness, and fatigue.

Shortness of breath and visual disturbances.

Pruritus (itching), especially after a hot shower.

Splenomegaly (enlarged spleen).

Tinnitus

Nose bleeding

Polycythemia Signs & Symptoms



Headaches



Pruritus



Visual Disturbances



Night Sweats



Plethora



Splenomegaly



Hypertension



Thrombosis

myhematology.com

Plethora in medicine refers to an excess of bodily fluid, typically blood (polycythemia), resulting in symptoms like a red complexion, congestion, or swelling

Symptoms of Increased viscosity

- Angina
- Dyspnea
- Hypoxia
- Bone and joint pain
- Thrombophlebitis
- Weight loss
- Chronic cough
- Night sweats and sleep disturbances
- Burning sensation over fingers or toes
- Splenomegaly and hepatomegaly
- Formation of blood clots in the blood vessels

Diagnosis

History and Physical examination

History usually includes questions living at high altitude, breathing difficulty, sleep disturbances and chronic cough

Blood Tests:-

The blood will show,

An increase in the number of red blood cells

Elevated hematocrit measurement

Elevated levels hemoglobin

Very low levels of erythropoietin

Erythropoietin very low levels of erythropoietin (EPO), a hormone that stimulates bone marrow to produce new red blood cells, because the excess red blood cells turn off the production of EPO by the kidneys (negative feedback)

- Bone marrow aspiration or biopsy
- If the examination of the bone marrow shows that it's producing higher number of blood cells. It may be a sign of polycythemia.
- Other diagnostic measures are
 - Chest x-ray
 - ECG
 - ECHO
 - O2 saturation

Management

Phlebotomy

- Drawing a certain amount of blood out of the veins is a procedure called phlebotomy .
- It is the 1st treatment option for people with polycythemia
- This reduces the number of blood cells and decreases your blood volume making it easier for the blood to function properly

Medications to decrease blood cells

□ For people with polycythemia who are not helped by phlebotomy alone. Medications such as hydroxyurea, to suppress the bone marrow's ability to produce blood cells may be used.

□ Interferon alpha may be used to stimulate the immune system to fight the overproduction of red blood cells.

□ Aspirin

100 mg daily to reduce thrombosis and cardiovascular risk, unless there is a specific contraindication

□ Therapy for reducing itching

Antihistamine may be used or ultraviolet light treatment to relieve the itching discomfort.

Complications

- Hemorrhage and thrombosis can complicate both primary and secondary cases of polycythemia, however the risk of progression to marrow fibrosis or leukemia is exclusive to PV.
- It is vital to ensure correct diagnosis so that appropriate therapy can
- be instituted.

THANK YOU